PRINCE VATS

Ph.D. Researcher in Environmental Engineering

Department of Civil Engineering • Indian Institute of Technology Gandhinagar

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RESEARCH FOCUS & CORE INTERESTS

Aerosol Science & Particle Dynamics: Particle number size distribution (PNSD) reconstruction from light-scattering measurements, aerosol microphysics, atmospheric transformation, and human respiratory exposure.
Environmental IoT Networks: Design, multi-pollutant sensing (PM_1, $PM_{2.5}$, PM_{10}, CO_2, CO, TVOCs), automated data telemetry, and ML-based field calibration against reference monitors.
Spatio-Temporal Deep Learning: Advanced neural network architectures (CNNs, GNNs, LSTMs, GRUs) for high-resolution dynamic forecasting, spatial air pollution interpolation, and pattern recognition.
Health Risk Modeling: Deterministic and stochastic environmental exposure analysis, indoor air quality mitigation via phytoremediation, and epidemiological health outcome assessments.

EDUCATION

Indian Institute of Technology Gandhinagar 2024 – Present (Ongoing)
Ph.D. Candidate, Environmental Engineering *Gandhinagar, Gujarat, India*

- **Doctoral Research:** AI-driven spatio-temporal analysis and high-resolution prediction of urban air quality using dense IoT-based environmental monitoring networks.
- Development and field deployment of low-cost Environmental Quality Monitoring Systems (EQMS) for high-frequency urban monitoring.
- Machine learning pipelines for field sensor calibration, data fusion, and exposure modeling.
- Deep learning models (GNN, LSTM/GRU, CNN-BiLSTM) for short-term and multi-scale $PM_{2.5}$ forecasting.

Indian Institute of Technology Ropar 2022 – 2024
Master of Technology (M.Tech), Environmental Engineering *Rupnagar, Punjab, India*

- **M.Tech Thesis:** Investigated indoor air pollution (PM_{10}, $PM_{2.5}$, PM_1, and black carbon) across classrooms, hostels, and offices; $PM_{2.5}$ (59.6 µg/m³) and PM_{10} (71.12 µg/m³) substantially exceeded CPCB limits.
- **Phytoremediation Strategy:** Tested Ajwain (*Trachyspermum ammi*, 39.3% $PM_{2.5}$ reduction) and Money plant (*Epipremnum aureum*, 38.7% PM_{10} reduction), establishing practical room-scale botanical mitigation recommendations.
- **Respiratory Health Assessment:** Conducted clinical spirometry on 303 students, identifying 35.97% restrictive patterns, 34.65% obstructive patterns, and 74.59% asthma prevalence linked to particulate exposure.

G. B. Pant University of Agriculture and Technology (GBPUAT) 2018 – 2022
Bachelor of Technology (B.Tech), Civil Engineering *Pantnagar, Uttarakhand, India*

HONORS & NATIONAL COMPETITIVE EXAMINATIONS

- **GATE 2025: All India Rank (AIR) 29** (99.56 percentile) nationwide in India's premier Graduate Aptitude Test in Engineering.
- **GATE 2026: All India Rank (AIR) 32** (99.56 percentile) nationwide.
- **Best Paper Award:** 7th National Water Seminar (7NWS), National Institute of Hydrology (NIH), Roorkee (August 2023).

PEER-REVIEWED PUBLICATIONS

JOURNAL ARTICLES

[1] Dhada, I., **Vats, P.**, and Samal, S. K. (2025). Potential Health Risk Estimation Through Deterministic and Stochastic Approaches for the Air Pollutants Found Inside Bottling Beverage Industries of Himachal Pradesh in India. *Discover Environment*, Springer, March 2025.
DOI: 10.1007/s44274-025-00221-x

REFEREED BOOK CHAPTERS & CONFERENCE PROCEEDINGS

[2] Dhada, I., **Vats, P.**, and Singh, B. (2025). Probable Health Risk of Pollutants and Noise Found Inside Bottling Beverage Plant of Himachal Pradesh, India. In: Shiva Nagendra, S.M., Ramamurthy, P.C., Vardoulakis, S., Tantrakarnapa, K., Ball, R.J. (eds) *Select Proceedings of the 8th Indian International Conference on Air Quality Management (IICAQM 2023)*. Lecture Notes in Civil Engineering, vol 582, pp. 219–238. Springer, Singapore. *(Oral Presentation at IICAQM 2023).*
DOI: 10.1007/978-981-96-2359-4_18

CONFERENCE PRESENTATIONS & SCHOLARLY CONTRIBUTIONS

PEER-REVIEWED ORAL PRESENTATIONS

[1] **Vats, P.**, Thakur, A. K., and Patel, S. (2025, Dec). "Development and Deployment of a Low-Cost IoT Based Environmental Quality Monitoring System (EQMS) with Spatio-Temporal Predictive Modeling", *14th Asian Aerosol Conference (AAC 2025)*, Indian Aerosol Science and Technology Association (IASTA), Mumbai, India. *(Presented by P. Vats).*

[2] Dhada, I., **Vats, P.**, and Singh, B. (2023, Dec). "Probable Health Risk of Pollutants and Noise Found Inside Bottling Beverage Plant of Himachal Pradesh, India", *8th Indian International Conference on Air Quality Management (IICAQM 2023)*, IISc Bengaluru, India. *(Presented by I. Dhada; Published in Springer LNCE, 2025).*

REFEREED POSTER PRESENTATIONS

[3] **Vats, P.**, Nagar, C., Sen, S., and Patel, S. (2026, Apr). "Low-Cost Environmental Quality Monitoring System for High-Resolution Spatiotemporal Urban Air Quality Monitoring and Multi-Scale Forecasting", *Industry Partnership Conclave (IPC 2026)*, IIT Gandhinagar, India.

[4] **Vats, P.**, Nagar, C., Sen, S., and Patel, S. (2026, Apr). "Environmental Quality Monitoring System Using Low-Cost Sensors for Urban Air Quality Monitoring and Multi-Scale Forecasting", *KPCSD Poster Competition on Sustainability*, Kiran C. Patel Centre for Sustainable Development, IIT Gandhinagar, India.

[5] Singh, A. P., **Vats, P.**, and Jain, M. K. (2025, Dec). "ML Model Evaluation for AQI Prediction in IGP Using Air Quality and Meteorological Features", *14th Asian Aerosol Conference (AAC 2025)*, Mumbai, India.

[6] **Vats, P.**, and Patel, S. (2025, Oct). "Environmental Quality Monitoring System Using Low-Cost Sensors for Urban Air Quality Monitoring and Multi-Scale Forecasting", *Chemical Engineering Symposium (CES 2025)*, IIT Gandhinagar & SVNIT, India.

[7] **Vats, P.**, Singh, A. P., and Patel, S. (2025, Aug). "Evaluating CNN-LSTM and GNN-GRU Hybrid Models for Enhanced Air Quality Prediction and Dynamic Forecasting", *India Clean Air Summit (ICAS 2025)*, CSTEP, Bengaluru, India.

[8] Singh, A. P., **Vats, P.**, and Jain, M. K. (2025, Aug). "ML Model Evaluation for AQI Prediction in Gangetic Plain Using Air Quality and Meteorological Features", *India Clean Air Summit (ICAS 2025)*, CSTEP, Bengaluru, India.

[9] **Two Paper Presentations** (2023, Aug). *7th National Water Seminar (7NWS)*, National Institute of Hydrology (NIH), Roorkee, India. *(Conferred Best Paper Award).*

TEACHING & ACADEMIC MENTORSHIP

Fundamentals of Aerosol Science — Teaching Assistant AY 2025–2026, Semester II

Indian Institute of Technology Gandhinagar • Advanced Undergraduate & Postgraduate Level

- Assisted in instruction of core aerosol physics and atmospheric chemistry: particle mechanics, dynamics, size distribution functions (PNSD), and atmospheric transformation processes.
- Conducted problem-solving tutorial sessions, guided students on atmospheric datasets, and instructed on measurement instrumentation including Scanning Mobility Particle Sizers (SMPS) and Optical Particle Counters (OPC).

ES 101: Engineering Graphics — Graduate Teaching Fellow AY 2025–2026 & AY 2026–2027

Indian Institute of Technology Gandhinagar • First-Year B.Tech (~50 Students per cohort)

- Taught classical engineering drafting: orthographic projections, isometric views, and sectioning of solids.
- Conducted weekly laboratory sessions instructing 3D parametric CAD modeling in **Autodesk Inventor** and **Autodesk Fusion 360**; evaluated design coursework and mentored student mechanical projects.

Introduction to Writing — Teaching Assistant AY 2024–2025, Semester I

Indian Institute of Technology Gandhinagar • First-Year B.Tech Foundation Course

- Mentored first-year undergraduate students in academic and technical communication, structured argumentation, critical reading, and peer-review workflows; provided detailed editorial critique on technical writing.

RESEARCH PROJECTS & FIELD STUDIES

Environmental Quality Monitoring System (EQMS) for Urban Air Quality IIT Gandhinagar

- Developed an integrated, low-cost multi-pollutant sensing network measuring PM_1, $PM_{2.5}$, PM_{10}, CO_2, CO, TVOCs, and ambient meteorology (temperature, relative humidity, barometric pressure).
- Engineered end-to-end data pipeline streaming real-time measurements across a dense urban grid to a centralized database server.
- Implemented CNN-BiLSTM architectures for anomaly detection and multi-scale air quality forecasting across 24-hour, 7-day, and 1-month horizons to generate actionable environmental policy insights.

Institutional Indoor Air Quality, Phytoremediation & Spirometry Assessment IIT Ropar

- Conducted longitudinal indoor air quality monitoring across classrooms, hostels, and offices, demonstrating high exposure risks.
- Benchmarked botanical mitigation efficacy of indoor plant species and administered clinical spirometry testing on 303 human subjects to establish correlations between particulate exposure and respiratory health.

Occupational Exposure & Health Risk Assessment in Beverage Bottling Plants Himachal Pradesh

- Monitored TVOCs, PM_{10}, $PM_{2.5}$, PM_1, black carbon (BC), and industrial noise across 12 manufacturing facilities during monsoon.
- Formulated deterministic and stochastic health risk models, identifying non-carcinogenic hazard ratios (0.52–0.80) and carcinogenic BC risks (5.8E-04 to 6.8E-04) to guide industrial ventilation recommendations.

ACADEMIC SERVICE & EVENT ORGANIZATION

- **Organizing Team Member:** Faculty Development Program (FDP) for ASEAN and Global South, IIT Gandhinagar (September 2026).
- **Organizing Team Member:** Industry Partnership Conclave (IPC 2026), IIT Gandhinagar (April 2026).
- **Organizing Team Member:** Symposium on Promoting Aerosol Education and Research (PAER 2026), IIT Gandhinagar (February 2026).
- **Organizing Team Member:** Symposium on Advances in Tools and Technologies to Quantify Health Impacts of Air Pollution (APHES 2025), IIT Gandhinagar (December 2025).

SCIENTIFIC SOFTWARE & OPEN RESEARCH TOOLS

- **Atmospheric Gas Unit Converter:** Online scientific utility for converting ambient gaseous concentration units (ppb, μg/m³) across user-defined temperature and atmospheric pressure regimes. (prince-vats.github.io/unit-converter)
- **Sensor Calibration Tool for Atmospheric Measurements:** Interactive web-based data calibration framework designed for calibrating low-cost environmental sensors against reference-grade instrumentation. (prince-vats.github.io/sensor-calibration)

TECHNICAL & COMPUTATIONAL TOOLKIT

Programming & Software:	Python, C, C++, MATLAB, R, MS Office Suite
AI & Deep Learning:	CNN, Graph Neural Networks (GNN), LSTM, GRU, Transformers / Predictors, TensorFlow, PyTorch, Feature Engineering (PCA)
Spatial & CAD Prototyping:	ArcGIS, QGIS, Autodesk Inventor, Autodesk Fusion 360
Specialized Domain Context:	IoT Sensor Design & Network Deployment, Environmental Data Calibration, Deterministic & Stochastic Health Risk Assessments, Real-Time Air Quality Monitoring ($PM_{2.5}$, PM_{10}, TVOCs)
Languages:	English (Fluent), Hindi (Fluent), Japanese (Basic)

SPECIALIZED ACADEMIC WORKSHOPS ATTENDED

- **Smart Sensors and Data Analytics for Air Quality Management Workshop:** Indian Institute of Technology Bombay (March 2025).
- **AI4Water: Advanced Data-Driven and Artificial Intelligence Tools in Water Resources Workshop:** Indian Institute of Technology Roorkee (August 2023).

LEADERSHIP & COMMUNITY ENGAGEMENT

- **Vice President:** Green Club — spearheaded environmental initiatives, clean-up drives, and campus sustainability projects.
- **Community Mentorship:** Taught underprivileged students through an NGO initiative.
- **Technical Seminars & Hackathons:** Organized university technical seminars and actively participated in environmental hackathons.
- **Campus Sustainability:** Volunteered in community clean-up drives and environmental tree plantation initiatives.